PenicillinX-Project with R2

How accurately can we predict beta-lactam cross reactivity based on molecular descriptors?

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Hutten et al.

Figure 1 is a reconstruction of the matrix detailed in Hutten et al[1] Below them are the various molecular matching algorithms and their respective matrices, the performance of each algorithm, as well as their combined performance matching against Hutten et al is displayed in a table at the end of the document. Matrices generated using R library complexheatmap[2,3]. *This matrix was reconstructed by manually transcribing data from the figure in the original paper, and though it was checked for consistency against the original, it can't be guaranteed that it is free from transcription error.*

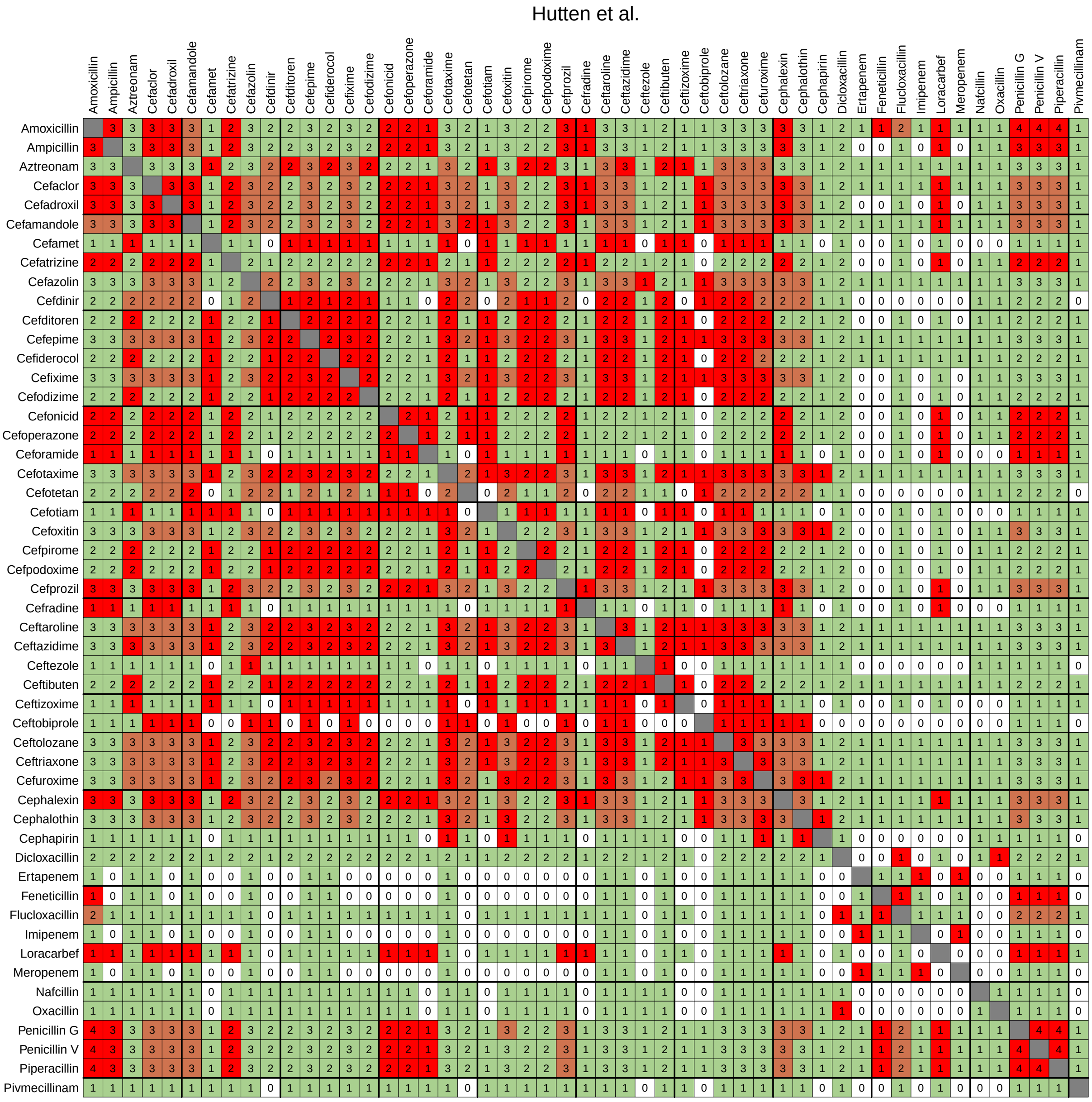

Figure 1: Cross reactivity matrix manually re-created from Hutten et al. Note: white:no data; Grey:cross-tabulation similar; Green:no crossreactivity; Safe; Orange:Cinically relevant discrepancy between studies; Red:(potential) crossreactivity: unsafe; N: nNumber of authors describing this potential cross reactivity as tabulated by Hutten et al

Direct Matching (PenicillinX Original)

The existing direct matching algorithm from PenicillinX web app.[4] Compares molecular structures directly using graph based comparisons of side chain SMILES[5]. It does not produce false positives, however sensitivity is limited, and often does not account for R1 alterations in side chain moieties that do not change allergenic potential.

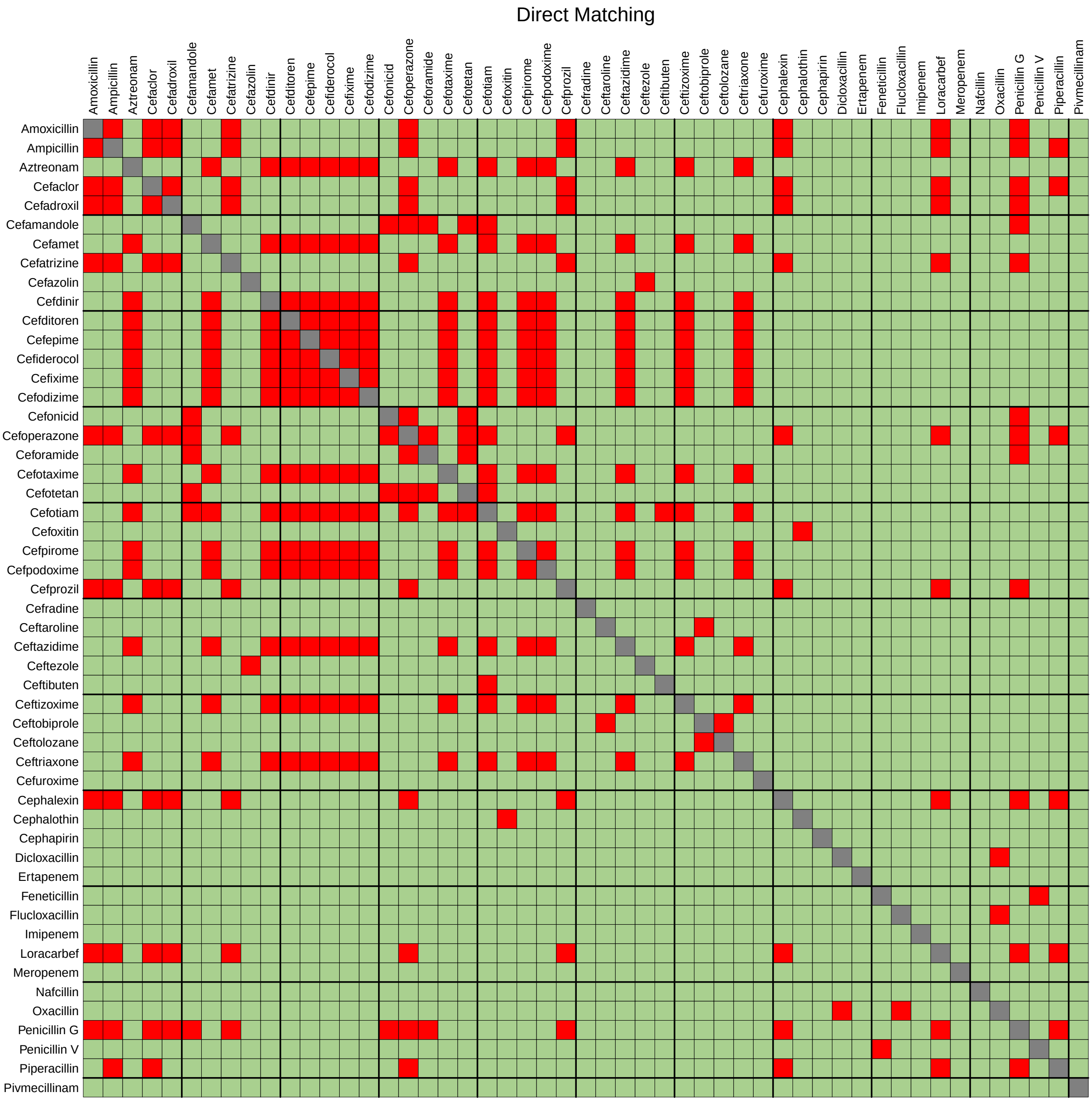

Figure 2: Cross reactivity matrix using PenicillinX direct matching algorithm

Permissive Direct Matching (PDM) using SMARTS substitutions

A Matching algorithm similar to original PenicillinX,[4] but instead using SMARTS pattern matching.[6] It implements bioisosteric substitution for features with similar expected behaviour specifically treating halogens as a united class (i.e. flucloxacillin-dicloxacillin) as well as methylene and ether linkages. (i.e. Penicillin G-VK). Continued work on this method shows promise in increasing the sensitivity of the algorithm while minimising loss to specificity.

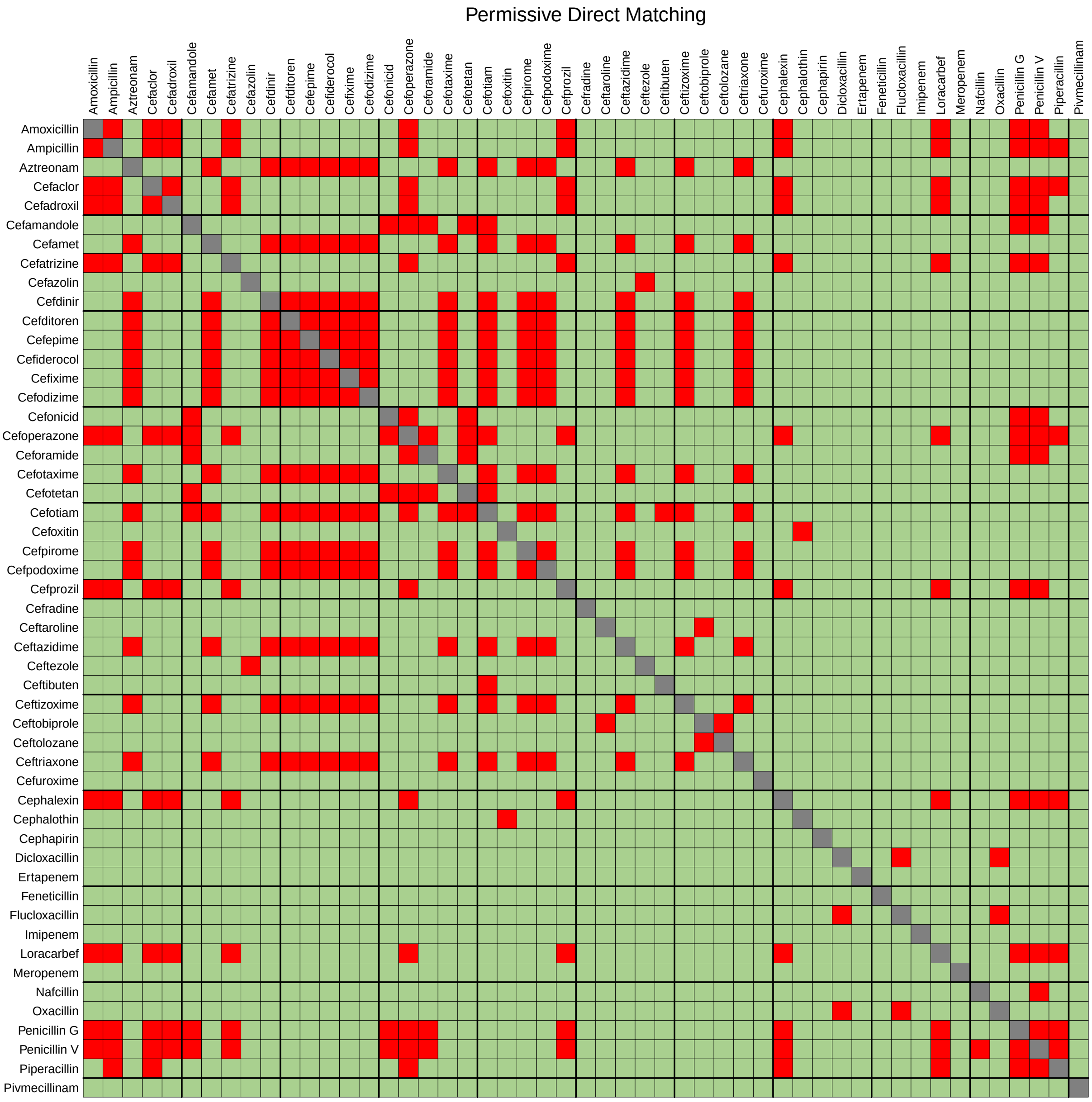

Figure 3: Cross reactivity matrix using an improved, more pemissive matching algorithm

Morgan Fingerprinting with Dice Coefficient

This method utilises Morgan circular fingerprints,[7] which iteratively encodes the local atomic environment into a bit vector. These fingerprints are then compared between two side chains using the Dice Coefficient[8,9]. The Dice coefficient assigns values to molecular structures, and weighs shared features more heaviliy than the total feature set, providing a measure of structural overlap.

$$S_{AB} = \frac{2C}{A + B} \tag{1}$$

where S_{AB} is similarity coefficient, C is the common features between both molecules, and A & B are the features in each molecule.

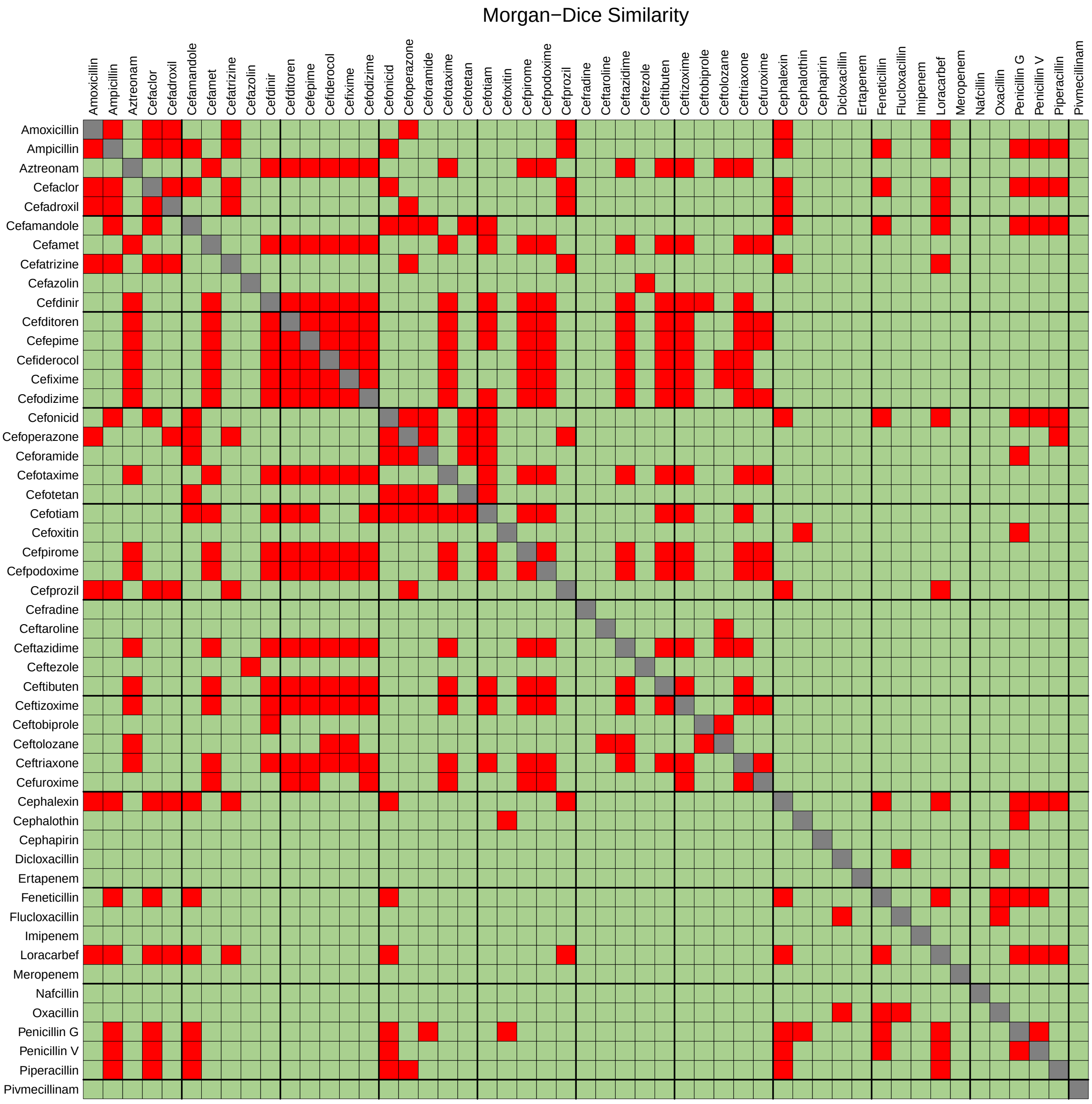

Figure 4: Cross reactivity matrix using Morgan fingerprinting with Dice Similarity

Morgan Fingerprinting with Tanimoto Coefficient

A variation of the previous algorithm that instead utilises the Tanimoto Coefficient[10]. Similar in concept to the Dice Coefficient, but with a different weight applied to matching substructures. Previous modelling in this project indicated that the Dice Coefficient strictly dominates the Tanimoto Coefficient, in detecting cross reactivity. Thus analyses using Tanimoto have been excluded from the final analysis.

$$S_{AB} = \frac{C}{A + B - C} \tag{2}$$

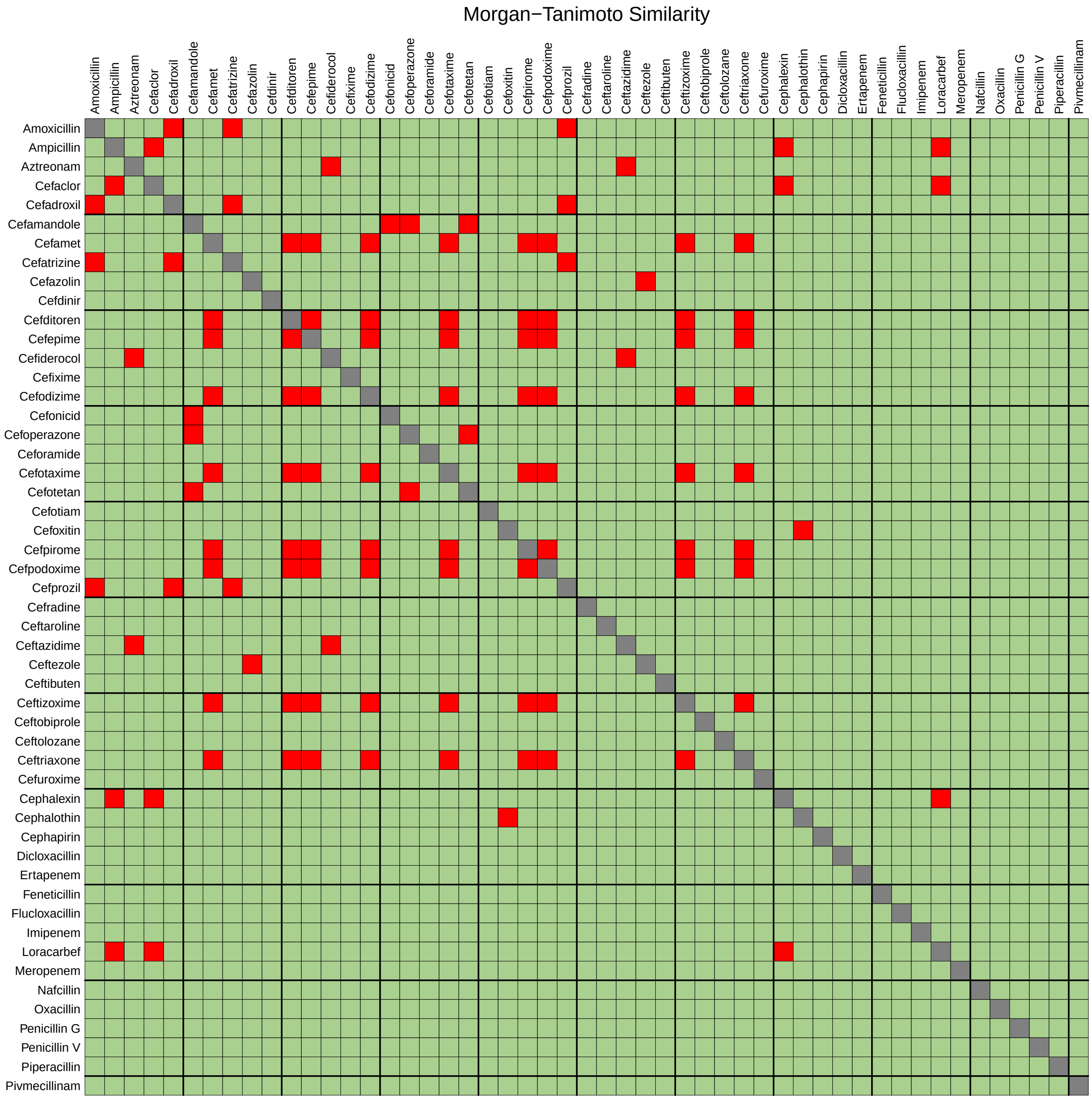

Figure 5: Cross reactivity matrix using Morgan fingerprinting with Tanimoto Similarity

Pharmacophore with Dice Coefficient

Pharmacophore fingerprinting maps the chemical characteristics of atoms and functional groups, such as hydrogen bond acceptors/donors, aliphatics/aromtatics. The Dice coefficient is then applied to attempt to model how an IgE molecule may 'perceive' a side chain's binding potential regardless of non-functional differences in the molecular structure.[9]

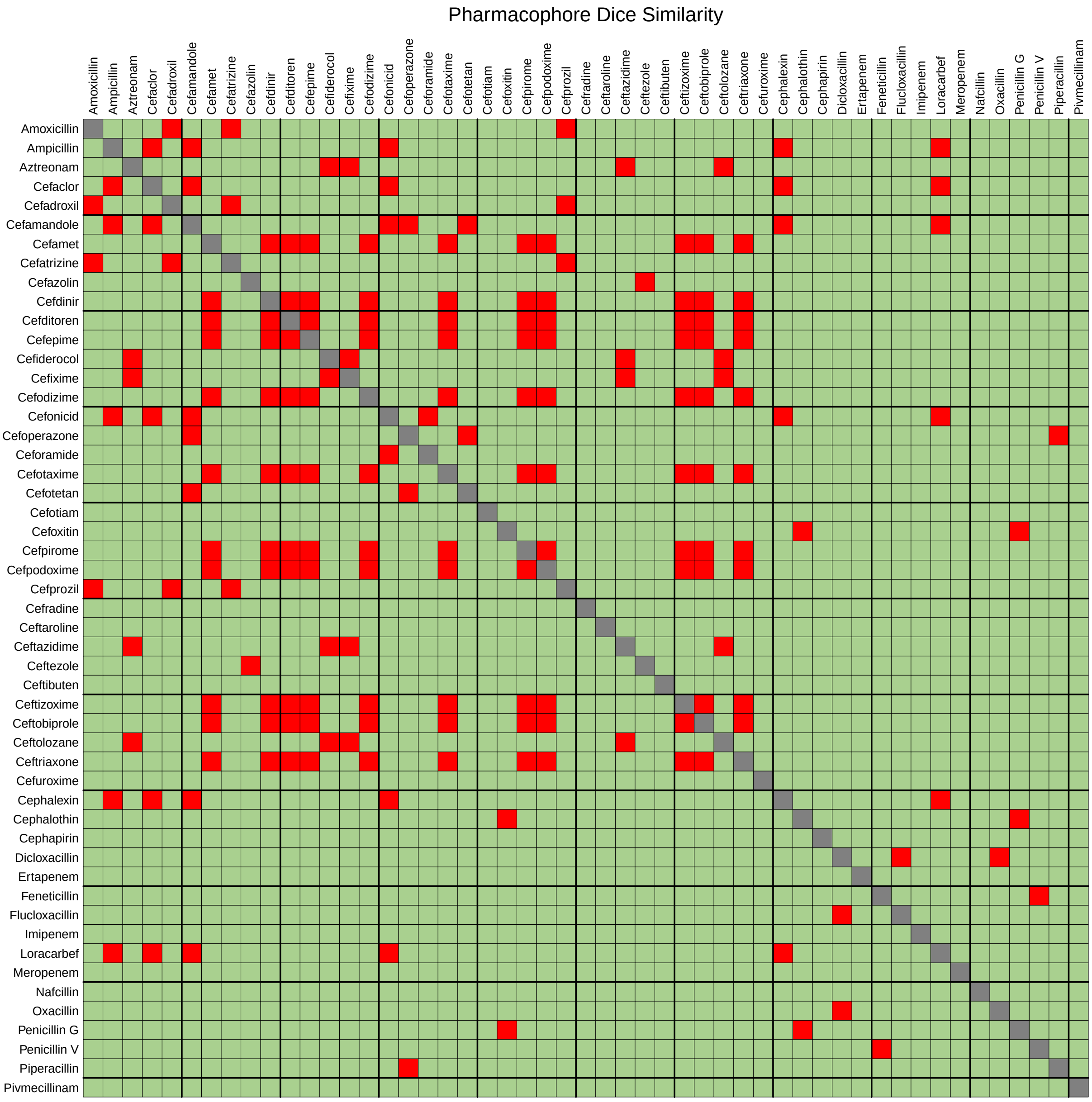

Figure 6: Cross reactivity matrix using Morgan fingerprinting with Pharmacophores

MACCS with Dice Coefficient

This method utilises Molecular Access System(MACCS) keys, a specific type of pre-defined, publicly available structural fragment definition. There are 166 public fragment definitions in the RDKit implementation. A bit string is generated with reference to the presence or absence of certain features, for example, < 3 oxygens, S-S bond, or ring of size 4, etc.[11,12] This method attempts to capture broad structural classes and motifs rather than atom-by-atom circular fingerprints. ;

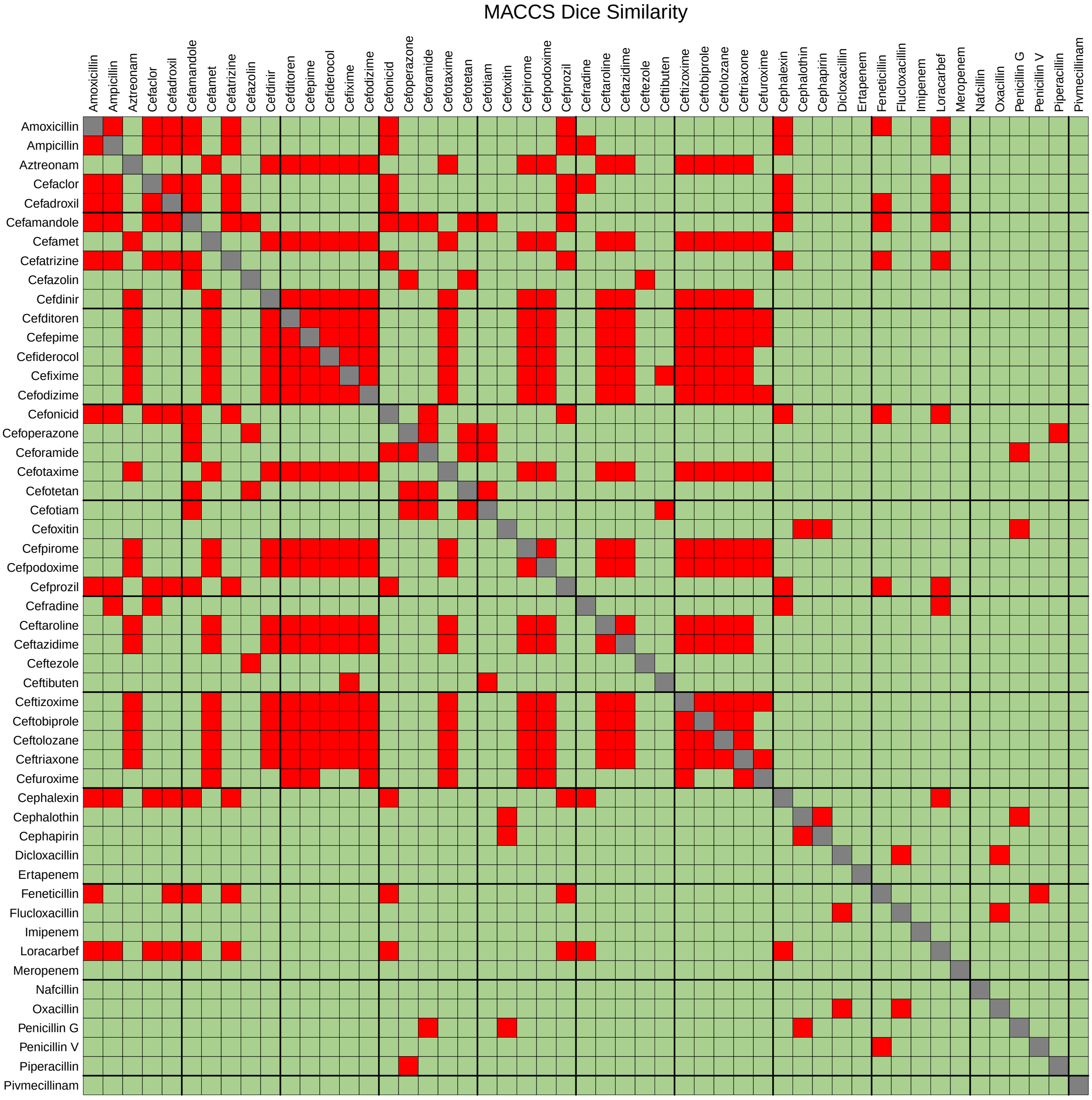

Figure 7: Cross reactivity matrix using MACCS keys with Dice Similarity

Maximum Common Substructure (MCS) Overlap

The MCS algorithm identifies the largest substructure shared by two side chains. It performs a graph-matching search to find the maximum number of identical atoms and bonds common to both molecules. A similarity score is calculated as the ratio of the shared atoms to the total atoms in the smaller of the two molecules.[13] Calculated as:

$$S_{MCS} = \frac{|MCS(A,B)|}{\min(|A|,|B|)} \tag{3}$$

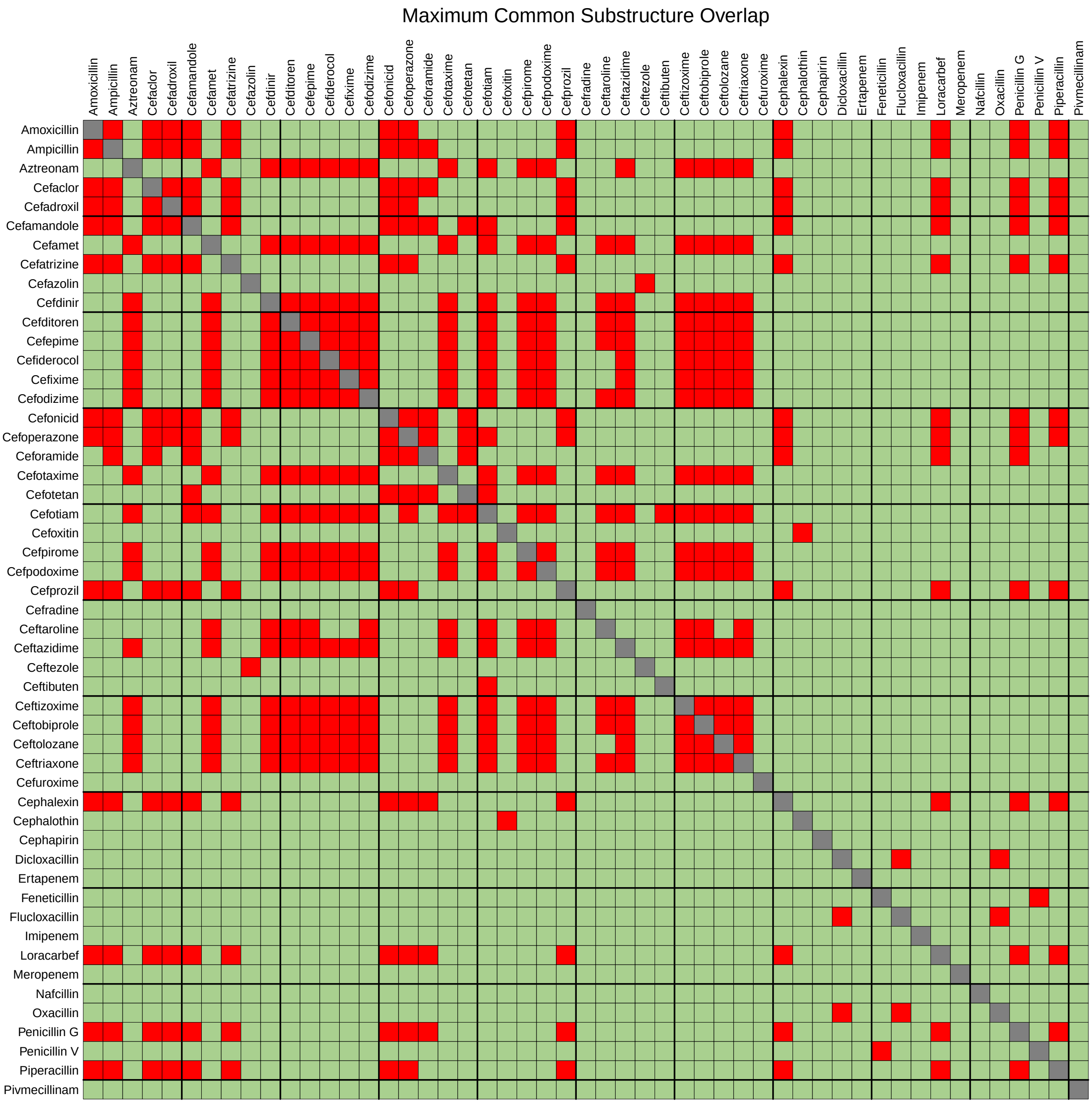

Figure 8: Cross reactivity matrix using Maximum Common Substructure

Atom Pairs With Dice Coefficient

The Atom Pairs descriptor[14], provides a topological representation based on relationships between every possible pair of non-hydrogen atoms within a given molecule. The descriptor records the chemical nature of each pair of atoms, comprising; atomic number, number of neighbouring atoms, and number of bonding pi-electrons. A value is assigned based on the shortest interatomic distance between the pair. The atomic properties and the interatomic distance combined are referred to as a 'triplet'. These are then aggregated into a vector, comprising a bit in the fingerprint. The atom pairs algorithm, with Tanimoto coefficient was used by Picard et al[15], using the now defunct ChemMine tool[16] to assign similarity scores to various β-lactam antibiotics, then applying these values along with empirical risk of cross reactivity in a linear regression analysis. Picard et al however, did not attempt to make formal predictions on the cross reactivity of particular β-lactams using defined similarity thresholds. Additionally, the Tanimoto Coefficient produces less robust results than using the Dice Coefficient.

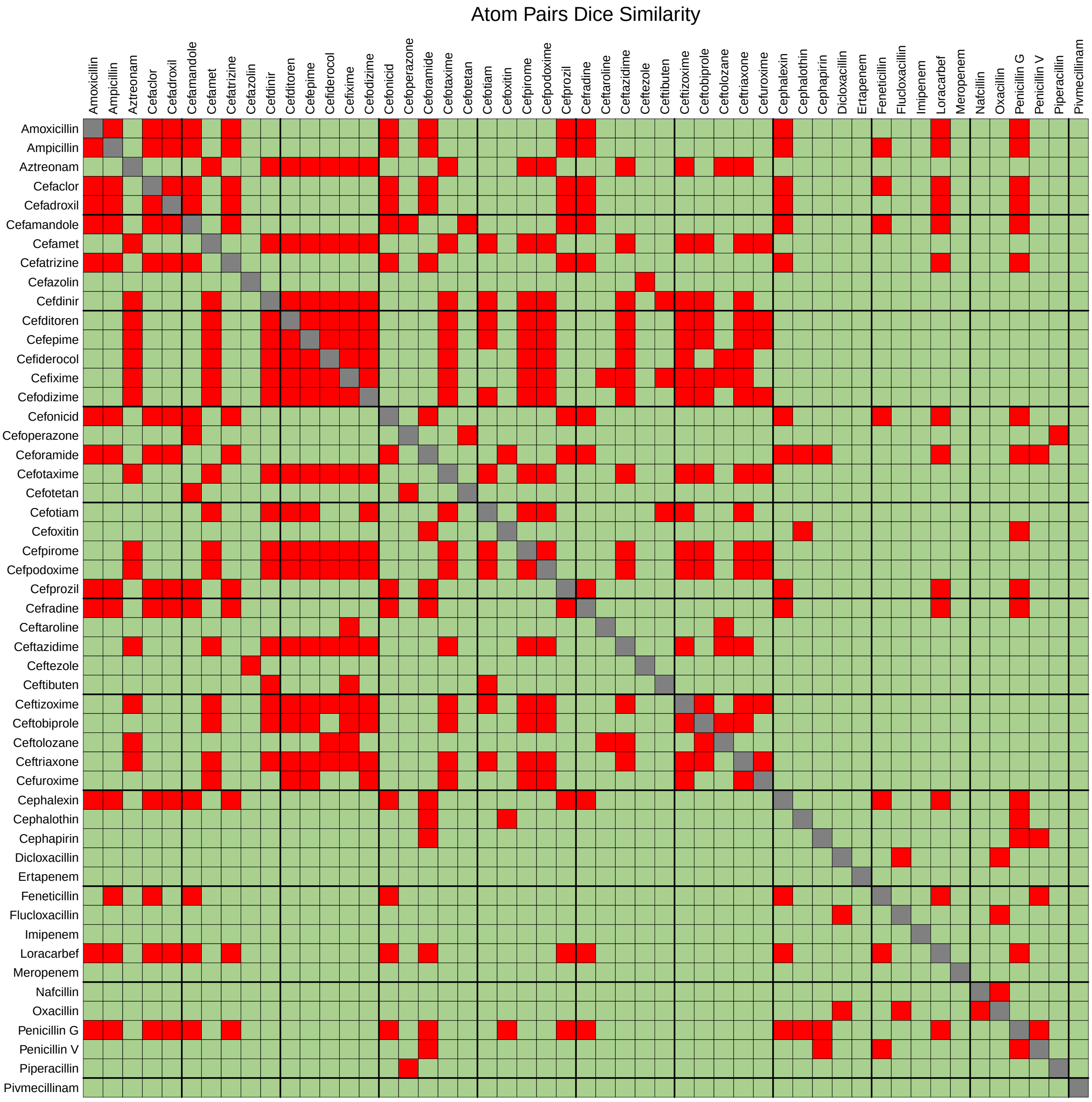

Figure 9: Cross reactivity matrix using Atom Pairs with Dice Similarity

All Algorithms Combined

A matrix generated with the combined hits of all previous algorithms, currently excluding Morgan Tanimoto due to poor performance (it does not find matches not already covered by other algorithms). MCS overlap has been re-added after further tweaking, having been removed in earlier iterations due to poor sensitivity. Atom-Pairs algorithm has been added. Cells are labelled with the number of individual algorithms indicating a potential cross reaction.

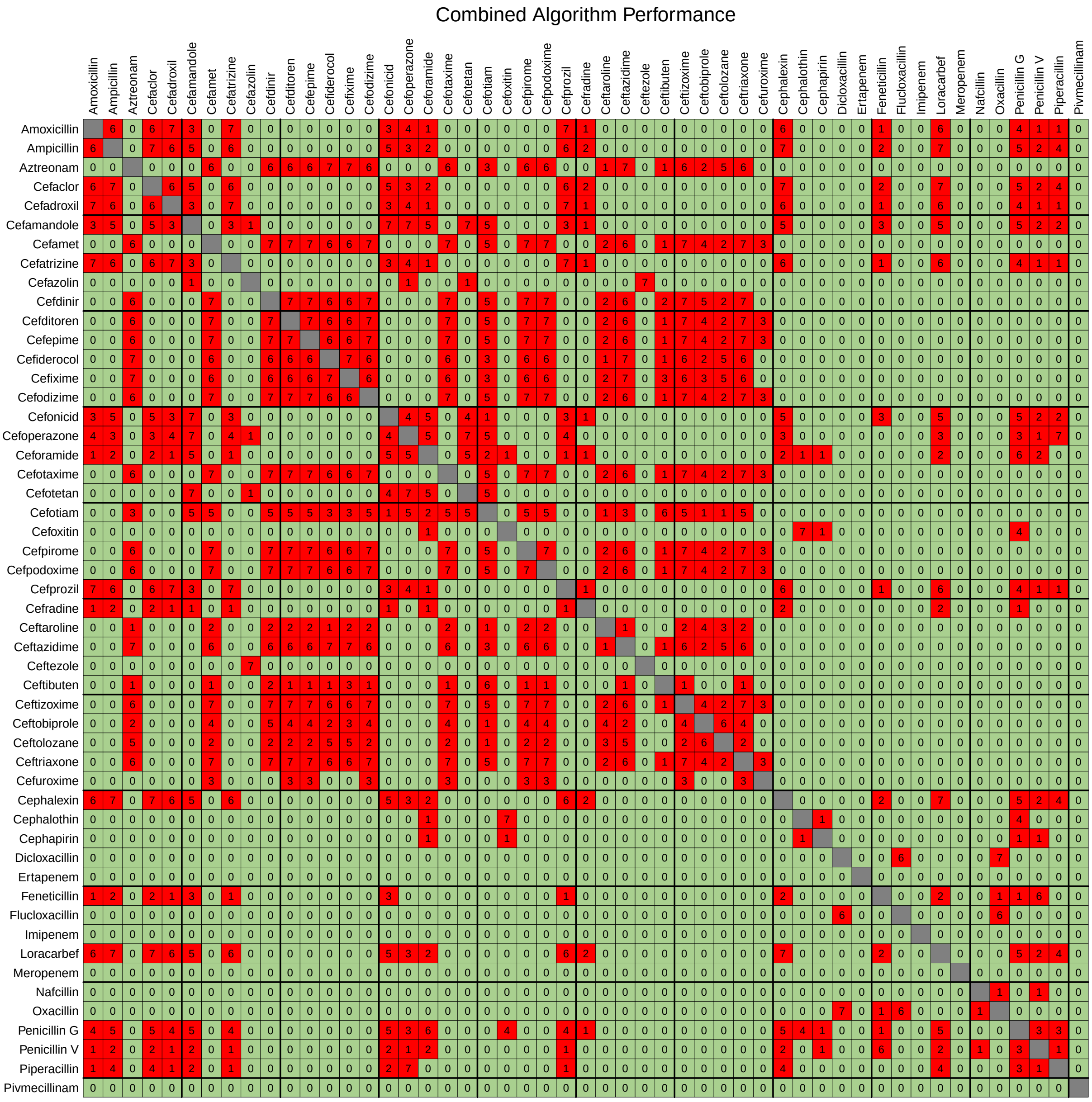

Figure 10: Combined performance of all algorithms with hit counts

Hutten et al. with Superimposed Algorithm Predictions

A re-creation of the Hutten et al[1] Matrix with black circles representing algorithmically predicted cross reactions seen in Figure 10.

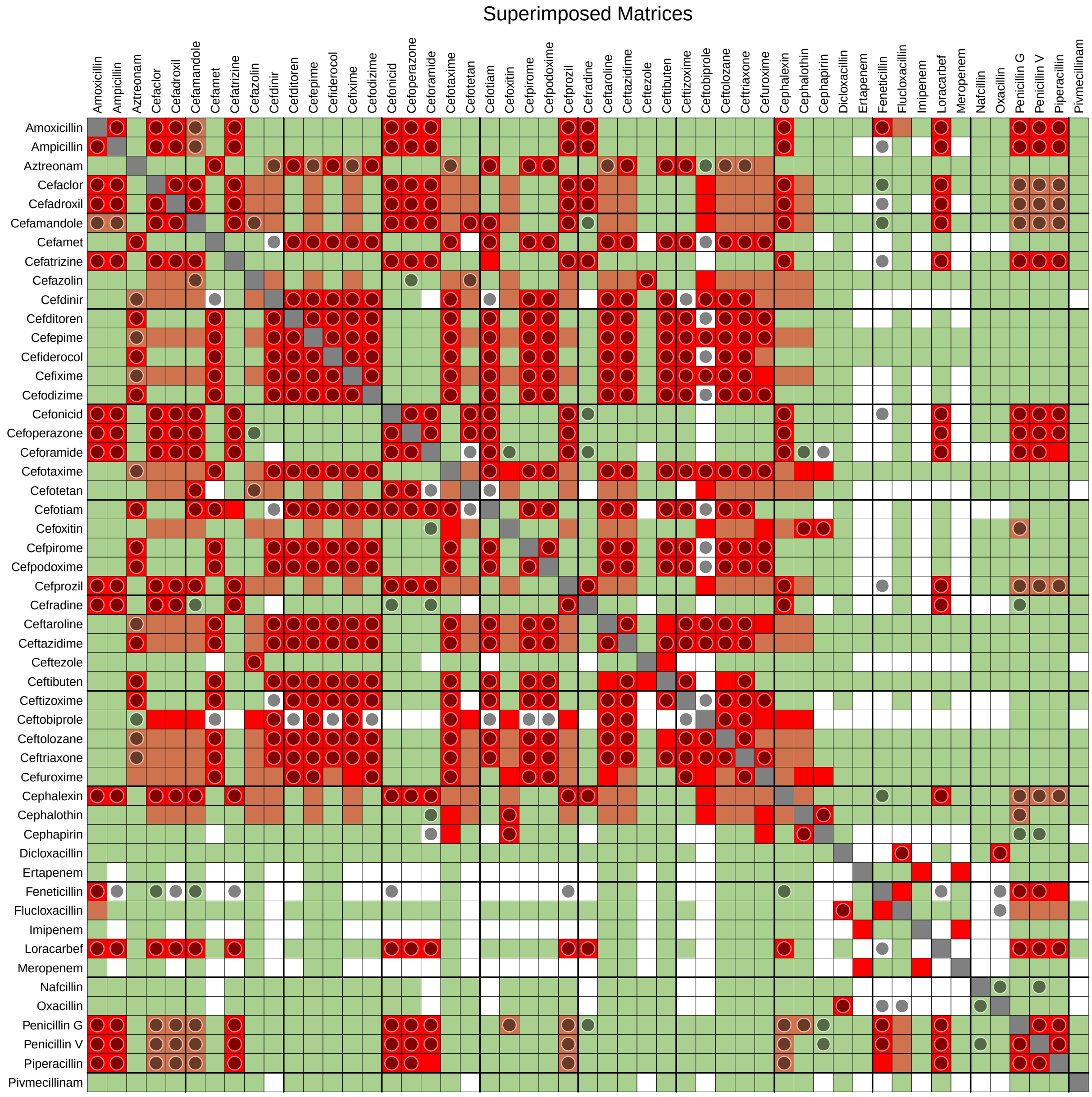

Figure 11: Algorithim predictions superimposed over Hutten et al. Matrix. Note: white:no data; Grey:cross-tabulation similar; Green:no crossreactivity; Safe; Orange:Cinically relevant discrepancy between studies; Red:(potential) crossreactivity: unsafe; Grey circle: Cross reactivity predicted by algorithm

Performance Table

A table detailing the performance of each algorithm with the final column demonstrating the performance of all algorithms combined. Youden's J is used as an overall performance metric to assist in gauging the cost/benefit of each change to an algorithm. Youden's J is calculated as:

$$J = \text{Sensitivity} + \text{Specificity} - 1 \tag{4}$$

I expect there should be a way that I can automate the process of tweaking the thresholds for the similarity coefficients to achieve an optimum Youden's J. Thus far these tweaks have been done manually, then revised by observing the result.

Table 1: Compared performance of each algorithm Morgan Tanimoto not included in combined result. DM:Direct Matching, MD:Morgan-Dice, PDM:Permissive Direct Matching, Pharmaco:Pharmacophore, MCS:Maximum Common Substructure, AP:Atom Pairs.

Metric	DM	MD	PDM	Pharmaco	MCS	MACCS	AP	Combo
Threshold	N/A	0.47983	N/A	0.671435	0.899807	0.774521	0.528055	N/A
Possible Hits	297.0	297	297.0	297	297	297	297	297.0
Total Hits	179.0	224	194.0	98	248	228	234	334.0
True +ve	161.0	193	170.0	85	215	196	193	269.0
False +ve	0.0	3	1.0	0	1	3	12	15.0
False -ve	136.0	104	127.0	212	82	101	104	28.0
no Data	6.0	10	6.0	8	14	16	14	22.0
Sensitivity	0.542	0.65	0.572	0.286	0.724	0.66	0.65	0.906
Specificity	1.0	0.995	0.998	1	0.998	0.995	0.982	0.977
Youden's J	0.542	0.645	0.571	0.286	0.722	0.655	0.632	0.883

Raw Results

A generated list of matches deemed to be false positives or negatives when compared to the Hutten et al. chart for auditing purposes. These can be plugged into PenicillinX[4] for visual inspection of the structures in order to generate hypotheses on why a particular algorithm may have incorrectly hit or missed potential interactions.

22 Predicted cross-reactions with no clinical data in Hutten et al.

- Ampicillin and Feneticillin [**MD, AP**]
- Cefadroxil and Feneticillin [**MACCS**]
- Cefamet and Cefdinir [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefatrizine and Feneticillin [**MACCS**]
- Cefdinir and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefdinir and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]

- Cefditoren and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefiderocol and Ceftobiprole [**MCS, MACCS**]
- Cefodizime and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefonicid and Feneticillin [**MD, MACCS, AP**]
- Ceforamide and Cefotetan [**DM, PDM, MD, MCS, MACCS**]
- Ceforamide and Cephapirin [**AP**]
- Cefotetan and Cefotiam [**DM, PDM, MD, MCS, MACCS**]
- Cefotiam and Ceftobiprole [**MCS**]
- Cefpirome and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefpodoxime and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefprozil and Feneticillin [**MACCS**]
- Ceftizoxime and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Feneticillin and Loracarbef [**MD, AP**]
- Feneticillin and Oxacillin [**MD**]
- Flucloxacillin and Oxacillin [**DM, PDM, MD, MCS, MACCS, AP**]

15 Potential false positives as compared to Hutten et al.

- Aztreonam and Ceftobiprole [**MCS, MACCS**]
- Cefaclor and Feneticillin [**MD, AP**]
- Cefamandole and Cefradine [**AP**]
- Cefamandole and Feneticillin [**MD, MACCS, AP**]
- Cefazolin and Cefoperazone [**MACCS**]
- Cefonicid and Cefradine [**AP**]
- Ceforamide and Cefoxitin [**AP**]
- Ceforamide and Cefradine [**AP**]
- Ceforamide and Cephalothin [**AP**]
- Cefradine and Penicillin G [**AP**]
- Cephalexin and Feneticillin [**MD, AP**]
- Cephapirin and Penicillin G [**AP**]
- Cephapirin and Penicillin V [**AP**]
- Nafcillin and Oxacillin [**AP**]
- Nafcillin and Penicillin V [**PDM**]

28 Predicted cross-reactions where there is discrepancy according to Hutten et al.

- Amoxicillin and Cefamandole [**MCS, MACCS, AP**]
- Ampicillin and Cefamandole [**MD, Pharmaco, MCS, MACCS, AP**]
- Aztreonam and Cefdinir [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Cefepime [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Cefixime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Aztreonam and Cefotaxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Ceftaroline [**MACCS**]
- Aztreonam and Ceftolozane [**MD, Pharmaco, MCS, MACCS, AP**]
- Aztreonam and Ceftriaxone [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefaclor and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Cefaclor and Penicillin V [**PDM, MD**]
- Cefaclor and Piperacillin [**DM, PDM, MD, MCS**]
- Cefadroxil and Penicillin G [**DM, PDM, MCS, AP**]
- Cefadroxil and Penicillin V [**PDM**]
- Cefadroxil and Piperacillin [**MCS**]
- Cefamandole and Cefazolin [**MACCS**]
- Cefamandole and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Cefamandole and Penicillin V [**PDM, MD**]
- Cefamandole and Piperacillin [**MD, MCS**]
- Cefazolin and Cefotetan [**MACCS**]
- Cefoxitin and Penicillin G [**MD, Pharmaco, MACCS, AP**]
- Cefprozil and Penicillin G [**DM, PDM, MCS, AP**]
- Cefprozil and Penicillin V [**PDM**]
- Cefprozil and Piperacillin [**MCS**]
- Cephalexin and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Cephalexin and Penicillin V [**PDM, MD**]
- Cephalexin and Piperacillin [**DM, PDM, MD, MCS**]
- Cephalothin and Penicillin G [**MD, Pharmaco, MACCS, AP**]

108 Negative predictions where there is discrepancy according to Hutten et al.

- Amoxicillin and Flucloxacillin [None]
- Aztreonam and Cefuroxime [None]
- Cefaclor and Cefazolin [None]
- Cefaclor and Cefdinir [None]
- Cefaclor and Cefepime [None]
- Cefaclor and Cefixime [None]
- Cefaclor and Cefotaxime [None]
- Cefaclor and Cefotetan [None]
- Cefaclor and Cefoxitin [None]
- Cefaclor and Ceftaroline [None]
- Cefaclor and Ceftazidime [None]
- Cefaclor and Ceftolozane [None]
- Cefaclor and Ceftriaxone [None]
- Cefaclor and Cefuroxime [None]
- Cefaclor and Cephalothin [None]
- Cefadroxil and Cefazolin [None]
- Cefadroxil and Cefdinir [None]
- Cefadroxil and Cefepime [None]
- Cefadroxil and Cefixime [None]
- Cefadroxil and Cefotaxime [None]
- Cefadroxil and Cefotetan [None]
- Cefadroxil and Cefoxitin [None]
- Cefadroxil and Ceftaroline [None]
- Cefadroxil and Ceftazidime [None]
- Cefadroxil and Ceftolozane [None]
- Cefadroxil and Ceftriaxone [None]
- Cefadroxil and Cefuroxime [None]
- Cefadroxil and Cephalothin [None]
- Cefamandole and Cefdinir [None]
- Cefamandole and Cefepime [None]
- Cefamandole and Cefixime [None]
- Cefamandole and Cefotaxime [None]
- Cefamandole and Cefoxitin [None]

- Cefamandole and Ceftaroline [None]
- Cefamandole and Ceftazidime [None]
- Cefamandole and Ceftolozane [None]
- Cefamandole and Ceftriaxone [None]
- Cefamandole and Cefuroxime [None]
- Cefamandole and Cephalothin [None]
- Cefazolin and Cefdinir [None]
- Cefazolin and Cefepime [None]
- Cefazolin and Cefixime [None]
- Cefazolin and Cefotaxime [None]
- Cefazolin and Cefoxitin [None]
- Cefazolin and Cefprozil [None]
- Cefazolin and Ceftaroline [None]
- Cefazolin and Ceftazidime [None]
- Cefazolin and Ceftolozane [None]
- Cefazolin and Ceftriaxone [None]
- Cefazolin and Cefuroxime [None]
- Cefazolin and Cephalexin [None]
- Cefazolin and Cephalothin [None]
- Cefdinir and Cefotetan [None]
- Cefdinir and Cefoxitin [None]
- Cefdinir and Cefprozil [None]
- Cefdinir and Cefuroxime [None]
- Cefdinir and Cephalexin [None]
- Cefdinir and Cephalothin [None]
- Cefepime and Cefotetan [None]
- Cefepime and Cefoxitin [None]
- Cefepime and Cefprozil [None]
- Cefepime and Cephalexin [None]
- Cefepime and Cephalothin [None]
- Cefiderocol and Cefuroxime [None]
- Cefixime and Cefotetan [None]
- Cefixime and Cefoxitin [None]
- Cefixime and Cefprozil [None]

- Cefixime and Cephalexin [None]
- Cefixime and Cephalothin [None]
- Cefotaxime and Cefotetan [None]
- Cefotaxime and Cefprozil [None]
- Cefotaxime and Cephalexin [None]
- Cefotetan and Cefoxitin [None]
- Cefotetan and Cefprozil [None]
- Cefotetan and Ceftaroline [None]
- Cefotetan and Ceftazidime [None]
- Cefotetan and Ceftolozane [None]
- Cefotetan and Ceftriaxone [None]
- Cefotetan and Cefuroxime [None]
- Cefotetan and Cephalexin [None]
- Cefotetan and Cephalothin [None]
- Cefoxitin and Cefprozil [None]
- Cefoxitin and Ceftaroline [None]
- Cefoxitin and Ceftazidime [None]
- Cefoxitin and Ceftolozane [None]
- Cefoxitin and Ceftriaxone [None]
- Cefoxitin and Cephalexin [None]
- Cefprozil and Ceftaroline [None]
- Cefprozil and Ceftazidime [None]
- Cefprozil and Ceftolozane [None]
- Cefprozil and Ceftriaxone [None]
- Cefprozil and Cefuroxime [None]
- Cefprozil and Cephalothin [None]
- Ceftaroline and Cephalexin [None]
- Ceftaroline and Cephalothin [None]
- Ceftazidime and Cefuroxime [None]
- Ceftazidime and Cephalexin [None]
- Ceftazidime and Cephalothin [None]
- Ceftolozane and Cefuroxime [None]
- Ceftolozane and Cephalexin [None]
- Ceftolozane and Cephalothin [None]

- Ceftriaxone and Cephalexin [None]
- Ceftriaxone and Cephalothin [None]
- Cefuroxime and Cephalexin [None]
- Cephalexin and Cephalothin [None]
- Flucloxacillin and Penicillin G [None]
- Flucloxacillin and Penicillin V [None]
- Flucloxacillin and Piperacillin [None]

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28 Potential false negatives as compared to Hutten et al.

- Cefaclor and Ceftobiprole [None]
- Cefadroxil and Ceftobiprole [None]
- Cefamandole and Ceftobiprole [None]
- Cefatrizine and Cefotiam [None]
- Cefazolin and Ceftobiprole [None]
- Cefixime and Cefuroxime [None]
- Ceforamide and Piperacillin [None]
- Cefotaxime and Cefoxitin [None]
- Cefotaxime and Cephalothin [None]
- Cefotaxime and Cephapirin [None]
- Cefotetan and Ceftobiprole [None]
- Cefoxitin and Ceftobiprole [None]
- Cefoxitin and Cefuroxime [None]
- Cefprozil and Ceftobiprole [None]
- Ceftaroline and Ceftibuten [None]
- Ceftaroline and Cefuroxime [None]
- Ceftezole and Ceftibuten [None]
- Ceftibuten and Ceftolozane [None]
- Ceftobiprole and Cefuroxime [None]
- Ceftobiprole and Cephalexin [None]
- Ceftobiprole and Cephalothin [None]
- Cefuroxime and Cephalothin [None]
- Cefuroxime and Cephapirin [None]
- Ertapenem and Imipenem [None]
- Ertapenem and Meropenem [None]

- Feneticillin and Flucloxacillin [None]
- Feneticillin and Piperacillin [None]
- Imipenem and Meropenem [None]

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269 True Positives in agreement with Hutten et al.

- Amoxicillin and Ampicillin [**DM, PDM, MD, MCS, MACCS, AP**]
- Amoxicillin and Cefaclor [**DM, PDM, MD, MCS, MACCS, AP**]
- Amoxicillin and Cefadroxil [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Amoxicillin and Cefatrizine [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Amoxicillin and Cefonicid [**MCS, MACCS, AP**]
- Amoxicillin and Cefoperazone *[DM, PDM, MD, MCS]*
- Amoxicillin and Ceforamide [**AP**]
- Amoxicillin and Cefprozil [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Amoxicillin and Cefradine [**AP**]
- Amoxicillin and Cephalexin [**DM, PDM, MD, MCS, MACCS, AP**]
- Amoxicillin and Feneticillin *[MACCS]*
- Amoxicillin and Loracarbef [**DM, PDM, MD, MCS, MACCS, AP**]
- Amoxicillin and Penicillin G [**DM, PDM, MCS, AP**]
- Amoxicillin and Penicillin V *[PDM]*
- Amoxicillin and Piperacillin *[MCS]*
- Ampicillin and Cefaclor [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Ampicillin and Cefadroxil [**DM, PDM, MD, MCS, MACCS, AP**]
- Ampicillin and Cefatrizine [**DM, PDM, MD, MCS, MACCS, AP**]
- Ampicillin and Cefonicid [**MD, Pharmaco, MCS, MACCS, AP**]
- Ampicillin and Cefoperazone *[DM, PDM, MCS]*
- Ampicillin and Ceforamide [**MCS, AP**]
- Ampicillin and Cefprozil [**DM, PDM, MD, MCS, MACCS, AP**]
- Ampicillin and Cefradine [**MACCS, AP**]
- Ampicillin and Cephalexin [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Ampicillin and Loracarbef [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Ampicillin and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Ampicillin and Penicillin V *[PDM, MD]*
- Ampicillin and Piperacillin *[DM, PDM, MD, MCS]*
- Aztreonam and Cefamet [**DM, PDM, MD, MCS, MACCS, AP**]

- Aztreonam and Cefditoren [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Cefiderocol [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Aztreonam and Cefodizime [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Cefotiam *[DM, PDM, MCS]*
- Aztreonam and Cefpirome [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Cefpodoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Aztreonam and Ceftazidime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Aztreonam and Ceftibuten *[MD]*
- Aztreonam and Ceftizoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefaclor and Cefadroxil [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefaclor and Cefamandole [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefaclor and Cefatrizine [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefaclor and Cefonicid [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefaclor and Cefoperazone *[DM, PDM, MCS]*
- Cefaclor and Ceforamide [**MCS, AP**]
- Cefaclor and Cefprozil [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefaclor and Cefradine [**MACCS, AP**]
- Cefaclor and Cephalexin [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefaclor and Loracarbef [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefadroxil and Cefamandole [**MCS, MACCS, AP**]
- Cefadroxil and Cefatrizine [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefadroxil and Cefonicid [**MCS, MACCS, AP**]
- Cefadroxil and Cefoperazone *[DM, PDM, MD, MCS]*
- Cefadroxil and Ceforamide [**AP**]
- Cefadroxil and Cefprozil [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefadroxil and Cefradine [**AP**]
- Cefadroxil and Cephalexin [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefadroxil and Loracarbef [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefamandole and Cefatrizine [**MCS, MACCS, AP**]
- Cefamandole and Cefonicid [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamandole and Cefoperazone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamandole and Ceforamide *[DM, PDM, MD, MCS, MACCS]*
- Cefamandole and Cefotetan [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamandole and Cefotiam *[DM, PDM, MD, MCS, MACCS]*

- Cefamandole and Cefprozil [**MCS, MACCS, AP**]
- Cefamandole and Cephalexin [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefamandole and Loracarbef [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefditoren [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefepime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefiderocol [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefamet and Cefixime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefamet and Cefodizime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefotaxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefamet and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Ceftaroline *[MCS, MACCS]*
- Cefamet and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefamet and Ceftibuten *[MD]*
- Cefamet and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Ceftolozane *[MCS, MACCS]*
- Cefamet and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefamet and Cefuroxime [**MD, MACCS, AP**]
- Cefatrizine and Cefonicid [**MCS, MACCS, AP**]
- Cefatrizine and Cefoperazone *[DM, PDM, MD, MCS]*
- Cefatrizine and Ceforamide [**AP**]
- Cefatrizine and Cefprozil [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefatrizine and Cefradine [**AP**]
- Cefatrizine and Cephalexin [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefatrizine and Loracarbef [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefatrizine and Penicillin G [**DM, PDM, MCS, AP**]
- Cefatrizine and Penicillin V *[PDM]*
- Cefatrizine and Piperacillin *[MCS]*
- Cefazolin and Ceftezole [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefditoren [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefepime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefiderocol [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefdinir and Cefixime [**DM, PDM, MD, MCS, MACCS, AP**]

- Cefdinir and Cefodizime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefotaxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Ceftaroline *[MCS, MACCS]*
- Cefdinir and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefdinir and Ceftibuten [**MD, AP**]
- Cefdinir and Ceftobiprole [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefdinir and Ceftolozane *[MCS, MACCS]*
- Cefdinir and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefepime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefiderocol [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefditoren and Cefixime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefditoren and Cefodizime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefotaxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefditoren and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Ceftaroline *[MCS, MACCS]*
- Cefditoren and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefditoren and Ceftibuten *[MD]*
- Cefditoren and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Ceftolozane *[MCS, MACCS]*
- Cefditoren and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefditoren and Cefuroxime [**MD, MACCS, AP**]
- Cefepime and Cefiderocol [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefepime and Cefixime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefepime and Cefodizime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Cefotaxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefepime and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Ceftaroline *[MCS, MACCS]*
- Cefepime and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]

- Cefepime and Ceftibuten *[MD]*
- Cefepime and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefepime and Ceftolozane *[MCS, MACCS]*
- Cefepime and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefepime and Cefuroxime [**MD, MACCS, AP**]
- Cefiderocol and Cefixime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefiderocol and Cefodizime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefiderocol and Cefotaxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefiderocol and Cefotiam *[DM, PDM, MCS]*
- Cefiderocol and Cefpirome [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefiderocol and Cefpodoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefiderocol and Ceftaroline *[MACCS]*
- Cefiderocol and Ceftazidime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefiderocol and Ceftibuten *[MD]*
- Cefiderocol and Ceftizoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefiderocol and Ceftolozane [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefiderocol and Ceftriaxone [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Cefodizime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Cefotaxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Cefotiam *[DM, PDM, MCS]*
- Cefixime and Cefpirome [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Cefpodoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Ceftaroline [**MACCS, AP**]
- Cefixime and Ceftazidime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefixime and Ceftibuten [**MD, MACCS, AP**]
- Cefixime and Ceftizoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefixime and Ceftobiprole [**MCS, MACCS, AP**]
- Cefixime and Ceftolozane [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefixime and Ceftriaxone [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefodizime and Cefotaxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefodizime and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefodizime and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefodizime and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]

- Cefodizime and Ceftaroline *[MCS, MACCS]*
- Cefodizime and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefodizime and Ceftibuten *[MD]*
- Cefodizime and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefodizime and Ceftolozane *[MCS, MACCS]*
- Cefodizime and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefodizime and Cefuroxime [**MD, MACCS, AP**]
- Cefonicid and Cefoperazone *[DM, PDM, MD, MCS]*
- Cefonicid and Ceforamide [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefonicid and Cefotetan *[DM, PDM, MD, MCS]*
- Cefonicid and Cefotiam *[MD]*
- Cefonicid and Cefprozil [**MCS, MACCS, AP**]
- Cefonicid and Cephalexin [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefonicid and Loracarbef [**MD, Pharmaco, MCS, MACCS, AP**]
- Cefonicid and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Cefonicid and Penicillin V *[PDM, MD]*
- Cefonicid and Piperacillin *[MD, MCS]*
- Cefoperazone and Ceforamide *[DM, PDM, MD, MCS, MACCS]*
- Cefoperazone and Cefotetan [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefoperazone and Cefotiam *[DM, PDM, MD, MCS, MACCS]*
- Cefoperazone and Cefprozil *[DM, PDM, MD, MCS]*
- Cefoperazone and Cephalexin *[DM, PDM, MCS]*
- Cefoperazone and Loracarbef *[DM, PDM, MCS]*
- Cefoperazone and Penicillin G *[DM, PDM, MCS]*
- Cefoperazone and Penicillin V *[PDM]*
- Cefoperazone and Piperacillin [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Ceforamide and Cefotiam *[MD, MACCS]*
- Ceforamide and Cefprozil [**AP**]
- Ceforamide and Cephalexin [**MCS, AP**]
- Ceforamide and Loracarbef [**MCS, AP**]
- Ceforamide and Penicillin G [**DM, PDM, MD, MCS, MACCS, AP**]
- Ceforamide and Penicillin V [**PDM, AP**]
- Cefotaxime and Cefotiam [**DM, PDM, MD, MCS, AP**]
- Cefotaxime and Cefpirome [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]

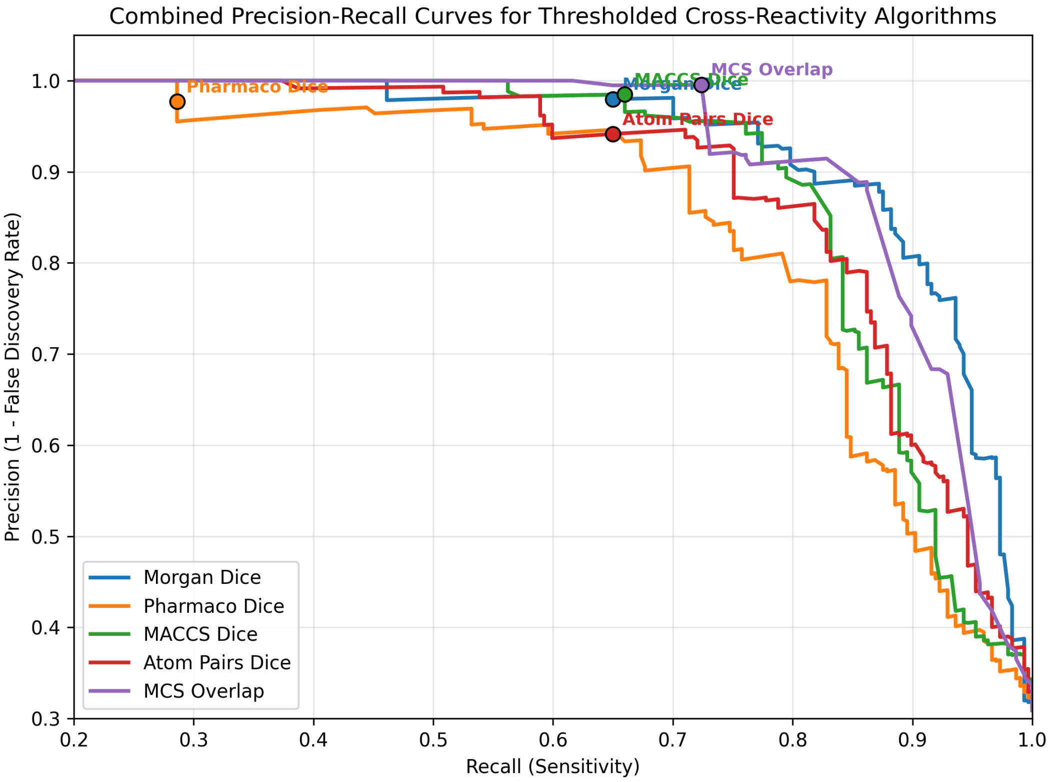
- Cefotaxime and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefotaxime and Ceftaroline *[MCS, MACCS]*
- Cefotaxime and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefotaxime and Ceftibuten *[MD]*
- Cefotaxime and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefotaxime and Ceftobiprole [**Pharmaco, MCS, MACCS, AP**]
- Cefotaxime and Ceftolozane *[MCS, MACCS]*
- Cefotaxime and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefotaxime and Cefuroxime [**MD, MACCS, AP**]
- Cefotiam and Cefpirome [**DM, PDM, MD, MCS, AP**]
- Cefotiam and Cefpodoxime [**DM, PDM, MD, MCS, AP**]
- Cefotiam and Ceftaroline *[MCS]*
- Cefotiam and Ceftazidime *[DM, PDM, MCS]*
- Cefotiam and Ceftibuten [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefotiam and Ceftizoxime [**DM, PDM, MD, MCS, AP**]
- Cefotiam and Ceftolozane *[MCS]*
- Cefotiam and Ceftriaxone [**DM, PDM, MD, MCS, AP**]
- Cefoxitin and Cephalothin [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefoxitin and Cephapirin *[MACCS]*
- Cefpirome and Cefpodoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefpirome and Ceftaroline *[MCS, MACCS]*
- Cefpirome and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefpirome and Ceftibuten *[MD]*
- Cefpirome and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefpirome and Ceftolozane *[MCS, MACCS]*
- Cefpirome and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefpirome and Cefuroxime [**MD, MACCS, AP**]
- Cefpodoxime and Ceftaroline *[MCS, MACCS]*
- Cefpodoxime and Ceftazidime [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefpodoxime and Ceftibuten *[MD]*
- Cefpodoxime and Ceftizoxime [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefpodoxime and Ceftolozane *[MCS, MACCS]*
- Cefpodoxime and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cefpodoxime and Cefuroxime [**MD, MACCS, AP**]

- Cefprozil and Cefradine [**AP**]
- Cefprozil and Cephalexin [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefprozil and Loracarbef [**DM, PDM, MD, MCS, MACCS, AP**]
- Cefradine and Cephalexin [**MACCS, AP**]
- Cefradine and Loracarbef [**MACCS, AP**]
- Ceftaroline and Ceftazidime *[MACCS]*
- Ceftaroline and Ceftizoxime *[MCS, MACCS]*
- Ceftaroline and Ceftobiprole *[DM, PDM, MCS, MACCS]*
- Ceftaroline and Ceftolozane [**MD, MACCS, AP**]
- Ceftaroline and Ceftriaxone *[MCS, MACCS]*
- Ceftazidime and Ceftibuten *[MD]*
- Ceftazidime and Ceftizoxime [**DM, PDM, MD, MCS, MACCS, AP**]
- Ceftazidime and Ceftobiprole *[MCS, MACCS]*
- Ceftazidime and Ceftolozane [**MD, Pharmaco, MCS, MACCS, AP**]
- Ceftazidime and Ceftriaxone [**DM, PDM, MD, MCS, MACCS, AP**]
- Ceftibuten and Ceftizoxime *[MD]*
- Ceftibuten and Ceftriaxone *[MD]*
- Ceftizoxime and Ceftolozane *[MCS, MACCS]*
- Ceftizoxime and Ceftriaxone [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Ceftizoxime and Cefuroxime [**MD, MACCS, AP**]
- Ceftobiprole and Ceftolozane [**DM, PDM, MD, MCS, MACCS, AP**]
- Ceftobiprole and Ceftriaxone [**Pharmaco, MCS, MACCS, AP**]
- Ceftolozane and Ceftriaxone *[MCS, MACCS]*
- Ceftriaxone and Cefuroxime [**MD, MACCS, AP**]
- Cephalexin and Loracarbef [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Cephalothin and Cephapirin *[MACCS]*
- Dicloxacillin and Flucloxacillin [**PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Dicloxacillin and Oxacillin [**DM, PDM, MD, Pharmaco, MCS, MACCS, AP**]
- Feneticillin and Penicillin G *[MD]*
- Feneticillin and Penicillin V [**DM, MD, Pharmaco, MCS, MACCS, AP**]
- Loracarbef and Penicillin G [**DM, PDM, MD, MCS, AP**]
- Loracarbef and Penicillin V *[PDM, MD]*
- Loracarbef and Piperacillin *[DM, PDM, MD, MCS]*
- Penicillin G and Penicillin V [**PDM, MD, AP**]

- Penicillin G and Piperacillin *[DM, PDM, MCS]*
- Penicillin V and Piperacillin *[PDM]*

PR Curve

(0.2, 1.0) (0.3, 1.05)



R2 Attributed Cross-Reactivity

The following pairs triggered a match exclusively because of their R2 side chains. For these pairs, the R1 side chains fell below the similarity threshold, but the R2 side chains exceeded it.

- Cefamandole and Cefazolin [MACCS] | **Discrepancy (Hutten: 2)**
- Cefamandole and Cefoperazone [PDM, Morgan, Pharmaco, MACCS, Atom Pairs] | **True +ve**
- Cefamandole and Ceforamide [PDM, Morgan, MACCS] | **True +ve**
- Cefamandole and Cefotetan [PDM, Morgan, Pharmaco, MACCS, Atom Pairs, MCS] | **True +ve**
- Cefamandole and Cefotiam [PDM, Morgan, MACCS, MCS] | **True +ve**
- Cefazolin and Cefoperazone [MACCS] | **False +ve**
- Cefazolin and Cefotetan [MACCS] | **Discrepancy (Hutten: 2)**
- Cefonicid and Cefoperazone [PDM, Morgan] | **True +ve**
- Cefonicid and Ceforamide [Morgan, Pharmaco, MACCS, Atom Pairs] | **True +ve**
- Cefonicid and Cefotetan [PDM, Morgan, MCS] | **True +ve**
- Cefonicid and Cefotiam [Morgan] | **True +ve**
- Cefoperazone and Ceforamide [PDM, Morgan, MACCS, MCS] | **True +ve**
- Cefoperazone and Cefotetan [PDM, Morgan, Pharmaco, MACCS, Atom Pairs, MCS] | **True +ve**
- Cefoperazone and Cefotiam [PDM, Morgan, MACCS, MCS] | **True +ve**
- Ceforamide and Cefotetan [PDM, Morgan, MACCS, MCS] | **No Data (Hutten: 3)**
- Ceforamide and Cefotiam [Morgan, MACCS] | **True +ve**
- Cefotetan and Cefotiam [PDM, Morgan, MACCS, MCS] | **No Data (Hutten: 3)**

Other studies

Table 2: Cross reference of OPCs results from other studies[17,18] with outcomes recorded in Hutten et al, as well as PenicillinX predicted result. Y:positive, N:Negative, D:Discrepant. Bracketed values correspond to number of included studies and number of algorithmic hits respectively

Reference	Index Abx	Challenge Abx	Study +ve	Hutten et al +ve	PX +ve
Touati et al 2021	Cefotaxime	Cefradine	Y	N(1)	N
	Cefotaxime	Cefalotin	Y	Y(3)	N
	Cefotaxime	Cefuroxime	Y	Y(3)	Y(3)
	Cefatrizine	Amoxicillin	Y	Y(2)	Y(7)
	Cefadroxil	Amoxicillin	Y	Y(3)	Y(7)
Stevensen et al 2025	Cefazolin	Cefalexin	N	D(3)	N
	Cefepime	Penicillin VK	N	N(3)	N

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